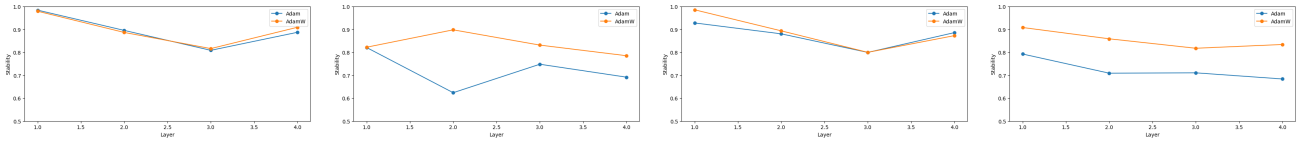
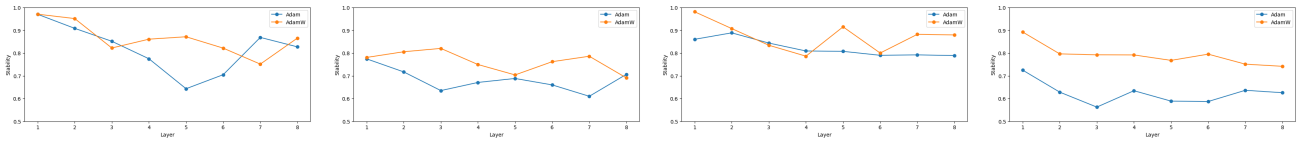
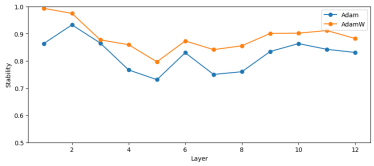


(a) Layers: 4, Heads: 8, Attn-only: False *(b)* Layers: 4, Heads: 8, Attn-only: True *(c)* Layers: 4, Heads: 16, Attn-only: False *(d)* Layers: 4, Heads: 16, Attn-only: True



(a) Layers: 8, Heads: 8, Attn-only: False *(b)* Layers: 8, Heads: 8, Attn-only: True *(c)* Layers: 8, Heads: 16, Attn-only: False *(d)* Layers: 8, Heads: 16, Attn-only: True



(a) Layers: 12, Heads: 12, Attn-only: False

B.6.1. MECHANISTIC CHECK

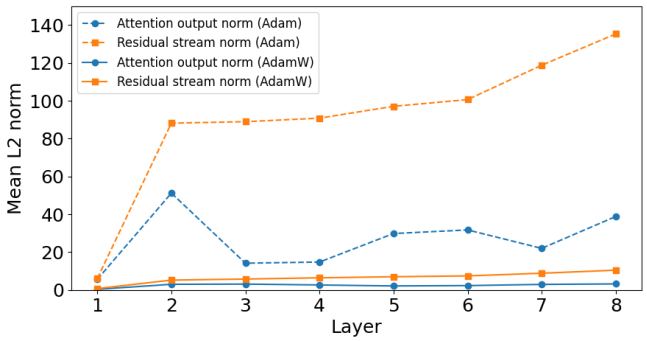
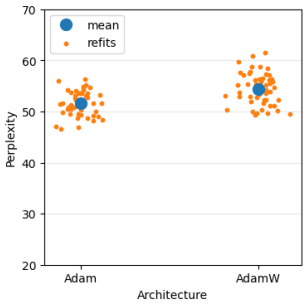


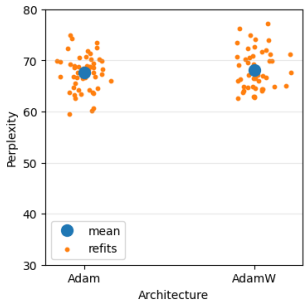
Figure 35. AdamW substantially controls norm growth. See § 4.5

B.7. Performance Parity: Adam vs AdamW

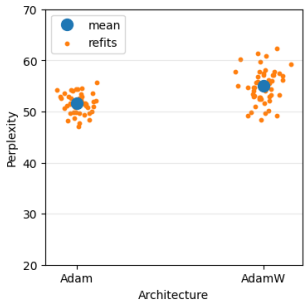
Additional architecture-wise plots showing **Performance Parity** between Adam and AdamW variants as mentioned in § 4.5. The validation perplexity was calculated using dataset ”wikitext-2-raw-v1” (Merity et al., 2016):



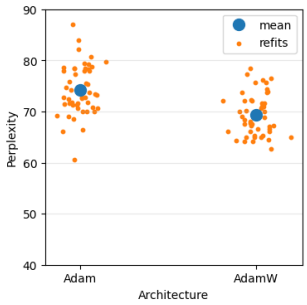
(a) Layers: 2, Heads: 8, Attn-only: False



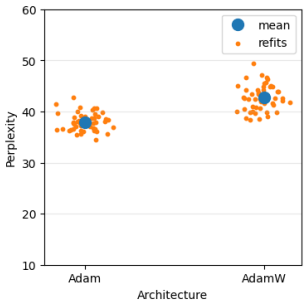
(b) Layers: 2, Heads: 8, Attn-only: True



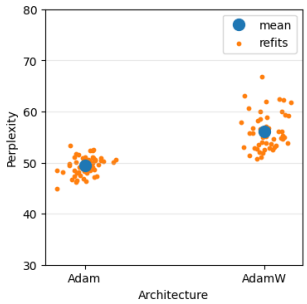
(c) Layers: 2, Heads: 16, Attn-only: False



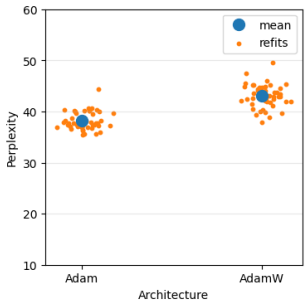
(d) Layers: 2, Heads: 16, Attn-only: True



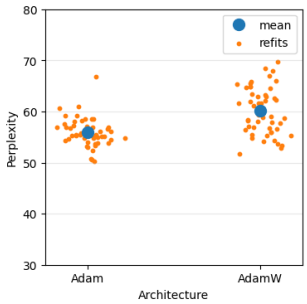
(a) Layers: 4, Heads: 8, Attn-only: False



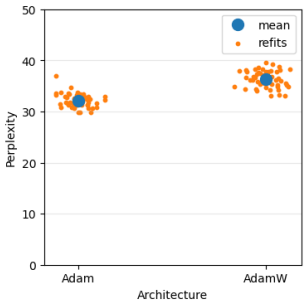
(b) Layers: 4, Heads: 8, Attn-only: True



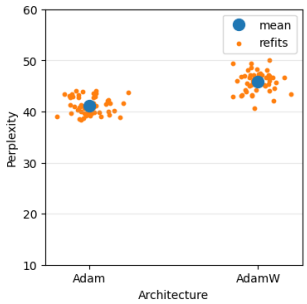
(c) Layers: 4, Heads: 16, Attn-only: False



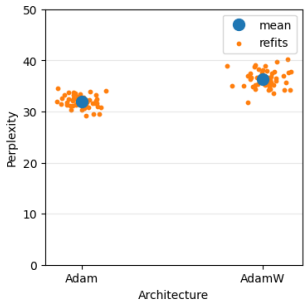
(d) Layers: 4, Heads: 16, Attn-only: True



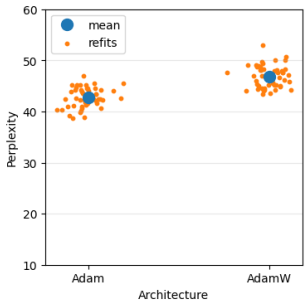
(a) Layers: 8, Heads: 8, Attn-only: False



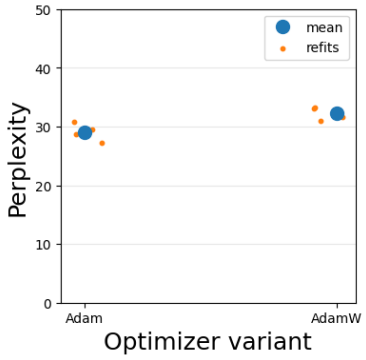
(b) Layers: 8, Heads: 8, Attn-only: True



(c) Layers: 8, Heads: 16, Attn-only: False



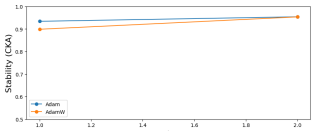
(d) Layers: 8, Heads: 16, Attn-only: True



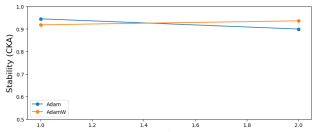
(a) Layers: 12, Heads: 12, Attn-only: False

B.8. Stability of residual stream

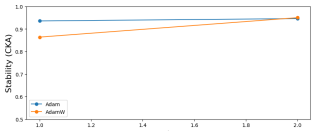
Additional architecture-wise plots for § 4.7:



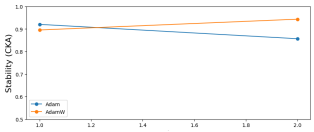
(a) Layers: 2, Heads: 8, Attn-only: False



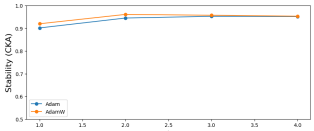
(b) Layers: 2, Heads: 8, Attn-only: True



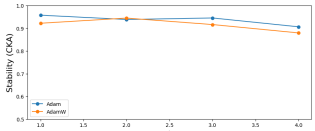
(c) Layers: 2, Heads: 16, Attn-only: False



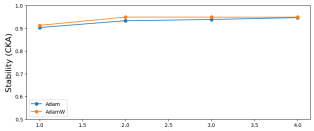
(d) Layers: 2, Heads: 16, Attn-only: True



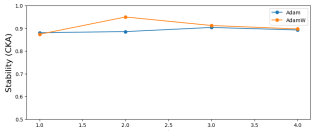
(a) Layers: 4, Heads: 8, Attn-only: False



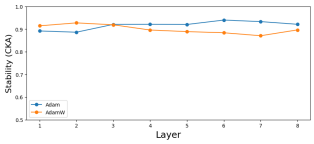
(b) Layers: 4, Heads: 8, Attn-only: True



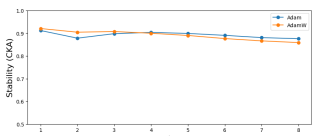
(c) Layers: 4, Heads: 16, Attn-only: False



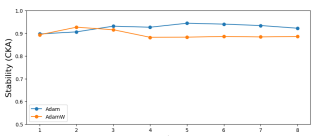
(d) Layers: 4, Heads: 16, Attn-only: True



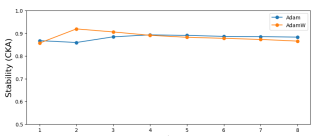
(a) Layers: 8, Heads: 8, Attn-only: False



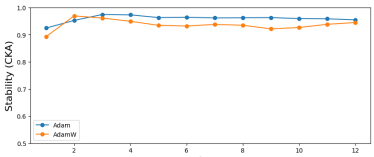
(b) Layers: 8, Heads: 8, Attn-only: True



(c) Layers: 8, Heads: 16, Attn-only: False



(d) Layers: 8, Heads: 16, Attn-only: True



(a) Layers: 12, Heads: 12, Attn-only: False

B.9. Correlation between the stability and "post-ablation change in perplexity"

Additional architecture-wise plots for § 4.8: